

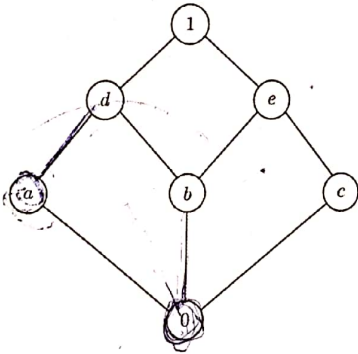
End-Term Examination – May 2026

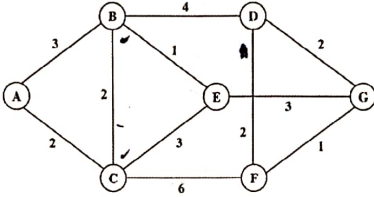
Subject Name / Code: Discrete Structure, CSE-2002

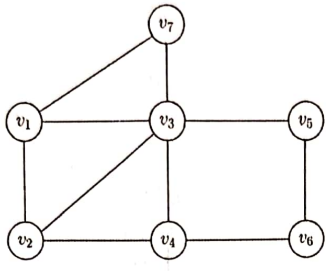
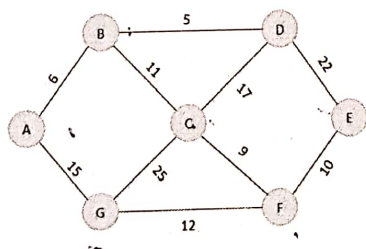
Max. Marks: 60

Time Allotted: 03 Hrs

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No.	Question	Marks
1.	(a) Consider the lattice $L = \{0, a, b, c, d, e, 1\}$ represented by the Hasse diagram:  (i) Find all complements of each element in L. (ii) Determine whether L is a complemented lattice. (iii) Identify elements that have more than one complement, if any.	2+1+2
	(b) Consider the group $G = (\mathbb{Z}_{11}^*, \cdot_{11})$ under multiplication modulo 11. It is given that $g \neq 2$ is a generator of this group. (i) Compute the sequence of powers $g^k \pmod{11}$ for $k = 1, 2, \dots, 10$ and present the results in a table. (ii) Using your table, find the value of x such that $2^x \equiv 9 \pmod{11}$. (iii) List out all generators of this group.	2+1+2
2.	(a) Solve $a_n = 4a_{n-1} - 4a_{n-2} + 2^n + 4n$, $a_0 = 1, a_1 = 4$.	5
	(b) Solve using generating functions: $a_n = 3a_{n-1} + n3^n$, $n \geq 1$, with initial condition $a_0 = 1$.	5

S. No.	Question	Marks
3.	(a) If the alarm rings, then the guard wakes up. If the guard wakes up, then the thief is caught. If the thief is not caught, then the alarm did not ring. The thief was not caught. Using rules of inference, analyze whether the guard woke up. Provide justification for each step.	5
	(b) Show that the following statement is a tautology without using a truth table: $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r).$ Use logical equivalences and clearly justify each step.	5
4.	(a) Consider the following weighted, undirected graph.  Use Dijkstra's algorithm to find the shortest path from A to all other vertices.	5
	(b) A cable inspection network in a data center is modeled as an undirected graph with vertices $V = \{A, B, C, D, E, F, G, H\}$ and edges: $A \leftrightarrow B, A \leftrightarrow D, B \leftrightarrow C, B \leftrightarrow D, B \leftrightarrow E,$ $C \leftrightarrow E, C \leftrightarrow F, D \leftrightarrow E, D \leftrightarrow G,$ $E \leftrightarrow F, E \leftrightarrow H, F \leftrightarrow G, F \leftrightarrow H, G \leftrightarrow H$ A technician must inspect each cable exactly once, starting from vertex C. (i) Find the degree of each vertex and identify vertices of odd degree. Determine whether an Eulerian path or circuit exists. Justify. (ii) Can the technician start at C and traverse all edges exactly once? Explain. (iii) If not, add the minimum number of edges to make such a traversal possible and specify them.	2+1+2

S. No.	Question	Marks
5.	(a) Consider the following graph G.  (i) Find a maximal matching in G. (iii) Decide whether G admits a perfect matching. Justify your answer. (iv) Construct a maximal matching that is not maximum, if possible.	2+1+2
	(b) Consider the graph G obtained from the complete graph K_5 by deleting one edge $e = uv$, and then adding a new vertex x which is connected to exactly three vertices of K_5, including both u and v. (i) Draw a planar embedding of the graph G, if it exists. (ii) Determine the number of vertices, edges, and faces of G. (iii) Verify Euler's formula for the planar embedding (if applicable).	2+2+1
6.	(a) Let $G = K_{3,2}$ be the complete bipartite graph. (i) Find the chromatic polynomial $P_G(k)$ of G. (ii) Hence determine the number of proper colorings of G using exactly 3 colors.	4+1
	(b) Consider the connected weighted graph G.  (i) Determine the total number of spanning trees of G. (ii) Apply Kruskal's Algorithm to find a Minimum Spanning Tree (MST) of G.	3+2

All The Best